GEOfinder

GSE105028, GSE280307, GSE101731, GSE154744, GSE209675, GSE267931, GSE173536

Experiment Type(s):

- [ ] Expression profiling by RT-PCR
- [ ] Expression profiling by array
- [ ] Expression profiling by high throughput sequencing
- [ ] Genome binding/occupancy profiling by genome tiling array

Species:

- [ ] Arabidopsis thaliana
- [ ] Bos taurus
- [ ] Caenorhabditis elegans
- [ ] Danio rerio
- [ ] Drosophila melanogaster
- [ ] Escherichia coli
- [ ] Gallus gallus
- [x] Homo sapiens

Number of Samples:

- [x] 1-10
- [x] 11-50
- [x] 51-100
- [x] 101-500
- [x] 501-1000
- [x] 1000+

Start Year:

End Year:

Submit

- [x] Genome binding/occupancy profiling by high throughput sequencing
- [ ] Genome variation profiling by SNP array
- [ ] Genome variation profiling by genome tiling array
- [ ] Methylation profiling by array
- [ ] Methylation profiling by genome tiling array
- [ ] Methylation profiling by high throughput sequencing
- [ ] Non-coding RNA profiling by array
- [ ] Non-coding RNA profiling by high throughput sequencing
- [x] Other

- [ ] Mus musculus
- [ ] Oryza sativa
- [ ] Rattus norvegicus
- [ ] Saccharomyces cerevisiae
- [ ] Schizosaccharomyces pombe
- [ ] Sus scrofa
- [ ] synthetic construct

Top results (up to 100):

GSE ID	Title	Summary
GSE209667	Transcriptional responses of cancer cells to heat shock-inducing stimuli involve amplification of robust HSF1 binding	Background. The Heat Shock Response (HSR) is an evolutionarily conserved mechanism that helps cells adapt to environmental stresses. Heat Shock Factor 1 (HSF1) is a master HSR transcription factor that is long known to bind at promoters of its target HSP genes to rapidly activate their transcription. However, recent genome-wide studies revealed HSF1 binding at thousands of sites outside of heat-shock responsive gene promoters, implying a broader role of HSF1 in HSR. Results. We asked how the widespread HSF1 binding is established and to what extent it may be related to transcription activation. To answer these questions, we examined responses of two different human cancer cell lines to two stimuli that induce HSR, exposure to 42C and inorganic arsenic at the ambient temperature, by manually integrating ChIP-sequencing against HSF1, RNA Pol II, H3K4Me3 with nascent RNA- and ATAC-sequencing datasets in cells before and during HSR induced by these two stimuli. We show that widespread HSF1 binding is a temperature-agnostic hallmark of HSR whose genome-wide

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		patterns combine cell type specificity with robustness to the nature of a stimulus and magnitude of activation. Mechanistically, HSR involves amplification of pre-existing low level HSF1 binding with few changes in accessible chromatin. HSF1 binding patterns show more stability than transcriptionally activated genes both between conditions and between cell lines. Conclusions. Comparing responses of distant cell lines to two HSR-inducing stimuli allowed us to define several features of HSR. Genome-wide HSF1 binding is a sensitive readout of HSR whose genome-wide patterns retain cell type identity across a broad dynamic range. The pre-existing genomic architecture is retained during HSR down to individual loci. Context-specific transcription involving common HSF1 binding favors a model of regulation by combinatory interplay of rigid transcription factor patterns.
GSE209685	Transcriptional responses of cancer cells to heat shock-inducing stimuli involve amplification of robust HSF1 binding	Background. The Heat Shock Response (HSR) is an evolutionarily conserved mechanism that helps cells adapt to environmental stresses. Heat Shock Factor 1 (HSF1) is a master HSR transcription factor that is long known to bind at

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		promoters of its target HSP genes to rapidly activate their transcription. However, recent genome-wide studies revealed HSF1 binding at thousands of sites outside of heat-shock responsive gene promoters, implying a broader role of HSF1 in HSR. Results. We asked how the widespread HSF1 binding is established and to what extent it may be related to transcription activation. To answer these questions, we examined responses of two different human cancer cell lines to two stimuli that induce HSR, exposure to 42C and inorganic arsenic at the ambient temperature, by manually integrating ChIP-sequencing against HSF1, RNA Pol II, H3K4Me3 with nascent RNA- and ATAC-sequencing datasets in cells before and during HSR induced by these two stimuli. We show that widespread HSF1 binding is a temperature-agnostic hallmark of HSR whose genome-wide patterns combine cell type specificity with robustness to the nature of a stimulus and magnitude of activation. Mechanistically, HSR involves amplification of pre-existing low level HSF1 binding with few changes in accessible chromatin. HSF1 binding

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		patterns show more stability than transcriptionally activated genes both between conditions and between cell lines. Conclusions. Comparing responses of distant cell lines to two HSR-inducing stimuli allowed us to define several features of HSR. Genome-wide HSF1 binding is a sensitive readout of HSR whose genome-wide patterns retain cell type identity across a broad dynamic range. The pre-existing genomic architecture is retained during HSR down to individual loci. Context-specific transcription involving common HSF1 binding favors a model of regulation by combinatory interplay of rigid transcription factor patterns.
GSE130758	Chromatin conformation remains stable upon extensive transcriptional changes driven by heat shock [K562]	Heat shock (HS) causes dramatic and rapid changes in transcription of thousands of genes in metazoans. We have used insitu Hi-C in Drosophila S2 and human K562 cells to determine the effects of HS upon chromatin conformation. Our study shows that topologically associated domains (TADs), compartments and distal regulatory interactions remain unchanged upon HS.
GSE127844	Heat-induced transcriptional memory	Precision run-on sequencing (PRO-seq) of nascent RNA synthesis across the human

GSE ID	Title	Summary
	profiled in human K562 erythroleukemia cells.	genome. The PRO-seq data was generated from human K562 erythroleukemia cells that were either unconditioned (unCond) or preconditioned with multiple heat shock exposures (preCond). After two day recovery from the preconditioning, during which the cells underwent two mitotic divisions, the cells were additionally subjected to a single heat shock to measure nascent transcription in response to heat shock in unconditioned versus preconditioned cells.
GSE280712	TSA-sequencing in untreated (UN), sodium arsenite (SA) and heat shock (HS) stressed HeLa cells	Mammalian cells activate diverse defense mechanisms in response to stresses. Nuclear stress bodies (nSBs) are transient membraneless organelles in primates that only form upon severe stresses. Little is known on how their formation contributes to cellular homeostasis or whether nSBs have any pathophysiological roles. We observed that the de novo formation of nSBs accompanies enlarged pericentromeric Sat DNA chromatin loci, prompting us to investigate whether such an acute alteration impacts regional chromatin accessibility and gene expression. To test this, we performed an improved

GSE ID	Title	Summary
		Tyramide Signal Amplification (TSA) mapping to identify genes proximal to newly formed nSBs.
GSE154746	Heat-induced transcriptional memory profiled in human K562 erythroleukemia cells [PRO-seq]	Precision run-on sequencing (PRO-seq) of nascent RNA synthesis across the human genome. The PRO-seq data was generated from human K562 erythroleukemia cells that were either unconditioned (unCond) or preconditioned with a single heat shock exposure (preC). After two day recovery from the preconditioning, during which the cells underwent two mitotic divisions, the cells were additionally subjected to a single heat shock to measure nascent transcription in response to heat shock in unconditioned versus preconditioned cells.
GSE89382	Genome-wide maps of histone H4 acetylation in human K562 cells in optimal growth conditions and upon heat stress	Genome-wide characterization of histone H4 acetylation in K562 cells in optimal growth conditions (NHS) and upon 30 min heat stress at 42C (HS30).
GSE111904	Osmostress induced changes of chromatin architecture and transcription in mammalian cells [RNA-Seq, HiC]	Osmostress causes chromatin compaction and simultaneously triggers a massive transcriptional response in eukaryotic cells that is required for adaptation

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		and cell survival. However, thus far it was unknown how changes of chromatin organization and ongoing DNA transcription are linked under hyper-osmotic conditions. In this study, we focused on elucidating the extend of chromatin reoganization upon hyper-osmotic shock, its reversibility and its link with active transcription in human T47D cells.
GSE90716	Hijacking of stress response machinery by oncogenes in acute leukaemia [ChIP-seq]	HSF1 orchestrates the heat shock response pathway. This pathway is co-opted in cancer and provides critical stress relief from oncogenic stress. HSF1 has a diverse occupancy signature depending on the cell type. In this study, we performed HSF1 ChIP-seq analysis using the human T-ALL cell line CUTLL1 and P12. These results revealed the HSF1 chromatin binding signature in CUTLL1 and P12 cells. MYC is a driving oncogene in T-ALL. The non-oncogene addiction pathways that act downstream of this transcription factor to support anabolic pathways are not well-understood. For this reason, we performed MYC ChIP-seq analysis using the human T-ALL cell line CUTLL1. These results revealed that HSF1 and HSF1 targets are

GSE ID	Title	Summary
		included in the MYC binding signature.

GSE ID	Title	Summary
GSE121182	Dynamic behavior of the epigenetic machinery upon thermal stress	Maintenance of the epigenome is of utmost importance to warrant appropriate cellular functioning Here we analyzed PRC1/2 function, in response to cellular stress. We observe that heat shock (HS) leads to nucleolar accumulation of CBX proteins, and are strongly immobilized in a 1,6-hexanediol insensitive manner, suggestive of a HS-induced liquid-to-solid phase transition of the nucleolus. CBX protein recovery from the nucleolus is dependent on heat shock protein (HSP) activity. LC-MS/MS analyses of nucleoli shows HS-induced nucleolar accumulation of PRC1/2 subunits, various chromatin regulators, HSPs, and the 26S proteasome. Finally, we find that HS leads to a strong reduction of PRC1/2 chromatin binding at target genes and loss of H2AK119ub and H3K27me3. Our findings demonstrate that thermal stress results in a rapid accumulation of epigenetic regulators in the nucleolus, and a concomitant loss of PRC1/2 complex chromatin binding and associated epimarks, underlining that environmental stress directly alters the epigenome.

GSE ID	Title	Summary
GSE111923	Osmostress induced changes of chromatin architecture and transcription in mammalian cells [ChIP-Seq]	Osmostress causes chromatin compaction and simultaneously triggers a massive transcriptional response in eukaryotic cells that is required for adaptation and cell survival. However, thus far it was unknown how changes of chromatin organization and ongoing DNA transcription are linked under hyper-osmotic conditions. In this study, we focused on elucidating the extent of chromatin reoganization upon hyper-osmotic shock, its reversibility and its link with active transcription in human T47D cells.
GSE166456	Genome-wide maps of YTHDC1 upon Heat-shock (HS) or in untreated conditions (UT) in HeLa cells	we used the technnology of ChIP-sequencing to specifically analyze the recruitment of YTHDC1 onto active genes during heat shock
GSE57398	Genome-wide characterization of HSF1 binding in human WT and FBXW7 KO colon cancer cells under basal conditions and upon heat shock	FBXW7 modulates stress response by post-translational modification of HSF1 | HSF1 orchestrates the heat-shock response upon exposure to heat stress and activates a transcriptional program vital for cancer cells. In this study we assayed for genome-wide localization of HSF1 enrichment in the HCT116 FBXW7 KO colon cells and their wild type counterpart in untreated cells and upon heat shock. These results revealed

GSE ID	Title	Summary
		that accumulation of nuclear HSF1 in FBXW7 KO cells results in rewiring of the HSF1 transcriptional program.
GSE130778	Chromatin conformation remains stable upon extensive transcriptional changes driven by heat shock	This SuperSeries is composed of the SubSeries listed below.
GSE194364	The NK cell stress response status modulates anti-tumor immunity [ChIP-seq]	The tumor microenvironment (TME) harbors numerous types of cellular stress that can activate Heat Shock Factor 1 (HSF1), a central regulator of stress response. HSF1 has been shown to directly regulate diverse gene programs beyond the classical heat shock response in a cell-type- and context-specific manner. Here, we uncover significant variability in HSF1 levels and activity between immune populations in patient tumors, with the lowest in natural killer (NK) cells. We demonstrate that accumulation of activated HSF1 in the TME dampens anti-tumor immunity through impairing NK cell cytotoxicity, which could be rescued by NK cells with homeostatic levels of HSF1. HSF1 stabilization results in altered chromatin accessibility and expression of surface receptors and signaling proteins involved in NK cell activation. We further

GSE ID	Title	Summary
		reveal direct occupancy of HSF1 at promoters of genes encoding mediators of NK cell cytotoxicity. Single-cell transcriptional profiling of the TME in the context of elevated HSF1 revealed an inverse relationship between the stress response status and NK cell effector function. Collectively, this work identifies a novel role of HSF1 in regulating the NK cell activation state and subsequent anti-tumor functionality.
GSE209687	Transcriptional responses of cancer cells to heat shock-inducing stimuli involve amplification of robust HSF1 binding	This SuperSeries is composed of the SubSeries listed below.
GSE89230	Nucleotide-resolution maps of nascent RNA-synthesis in human K562 cells in optimal growth conditions and upon a 30-minute heat shock at 42C	Precision Run-On Sequencing (PRO-seq) mapping of nascent RNA synthesis at a nucleotide-resolution across the human genome. The PRO-seq data has been generated from human K562 cells that have been cultured in optimal growth conditions (NHS) or that have been exposed to an acute, 30-minute at 42C, heat treatment (HS30). The data contains two replicates in each condition (fastq files of raw sequencing data), and bigWig files where the two replicates have been combined and the

GSE ID	Title	Summary
		data normalized to allow direct comparison of the transcription profiles. in NHS and HS30 conditions.
GSE43579	Genome-wide maps of HSF1 and HSF2 binding sites in cycling and mitotic human K562 cells in optimal growth conditions and upon heat stress	Genome-wide characterization of binding sites for heat shock transcription factors HSF1 and HSF2 in freely cycling and mitotic K562 cells in optimal growth conditions and upon 30 min heat stress at 42C
GSE184139	Chromatin Landscape Analysis of H1 cells with different temperature.	Chromatin Landscape Analysis of H1 cells with different temperature.
GSE60984	Subsets of NFkappaB-dependent genes are differentially regulated after combined treatment with hyperthermia and cytokine	Heat shock transcription factor 1 (HSF1) is responsible for triggering stress-induced activation of the HSP genes. HSF1 is also involved in the regulation of many other genes associated with multiple cellular processes including cell signaling, development, fertility, cell death and metabolism, e.g. NFkappaB transcription factor. Here we aimed to establish a global picture for the association between hyperthermia-modulated expression of NFkappaB-dependent genes and HSF1 binding in regulatory regions of target genes. The global pattern of HSF1 binding was analyzed after 10 and 20 minutes of hyperthermia using

GSE ID	Title	Summary
		the ChIP-Seq approach. The analysis revealed about 25,000 HSF1 binding sites in chromatin of control untreated U-2 OS cells, which nearly doubled in cells subjected to hyperthermia. The presence of hyperthermia-induced HSF1 binding was established in regulatory regions of hyperthermia-affected genes and in TNF-affected genes. In most cases hyperthermia pre-conditioning inhibited cytokine-mediated activation of NF-B-dependent genes. However, our results also revealed novel gene subsets that included TNFalpha-regulated and hyperthermia-unaffected genes, as well as genes (co)repressed, (co)stimulated, or antagonized by both types of stimuli.
GSE253873	Histone H3.3 chaperone HIRA is required for acquired tolerance by rendering stress-responsive genes poised for prospective lethal stresses [ChIP-seq]	Appropriate responses to environmental challenges are imperative for the survival of all living organisms. Exposure to low-dose stresses is recognized to yield increased cellular fitness, a phenomenon termed hormesis. However, our molecular understanding of how cells respond to low-dose stress remains profoundly limited. Here we report that histone variant H3.3-specific chaperone, HIRA, is required for acquired tolerance, where low-dose heat stress exposure

GSE ID	Title	Summary
		confers resistance to subsequent lethal heat stress. We found that human HIRA activates stress-responsive genes, including HSP70, by depositing histone H3.3 following low-dose stresses. These genes are also marked with histone H3 Lys-4 trimethylation and H3 Lys-9 acetylation, both active chromatin markers. Moreover, depletion of HIRA greatly diminished acquired tolerance, both in normal diploid fibroblasts and in HeLa cells. Collectively, our study revealed that HIRA is required for eliciting adaptive stress responses under environmental fluctuations and is a master regulator of stress tolerance.

GSE ID	Title	Summary
GSE66448	Heat shock-induced changes in chromatin binding of SUMO2/3 modified proteins	Heat shock induces rapid modification of proteins with SUMO2/3. This study concentrated in charaterizing how these changes are reflected on SUMOylation of chromatin bound proteins, trancsription, and chromatin binding of SUMO ligase PIAS1.
GSE130892	MeRIP-seq for heat shock in B-cell lymphoma cells	We ran MeRIP-seq on RNA from heat shocked (43 C for 1 hr) and untreated cells to compare to previous reports as part of an evaluation of MeRIP-seq reproducibility.
GSE194098	HSF2 cooperates with HSF1 to drive a transcriptional program critical for the malignant state	Heat shock factor 1 (HSF1) is well known for its role in the heat shock response (HSR), where it drives a transcriptional program comprising heat shock protein (HSP) genes, and in tumorigenesis, where it drives a program comprising HSPs and many non-canonical target genes that support malignancy. Here, we find that HSF2, an HSF1 paralog with no significant role in the HSR, physically and functionally interacts with HSF1 across

GSE ID	Title	Summary
		diverse types of cancer. HSF1 and HSF2 have strikingly similar chromatin occupancy and regulate a common set of genes which include both HSPs and non-canonical transcriptional targets with roles critical in supporting malignancy. Loss of either HSF1 or HSF2 results in a dysregulated response to nutrient stresses in vitro and reduced tumor progression in cancer cell line xenografts. Taken together, these findings establish HSF2 as a critical cofactor of HSF1 in driving a cancer cell transcriptional program to support the anabolic malignant state.
GSE310047	Hyperosmotic stress induces a large-scale rewiring of 3D chromatin interactions [Cut & Tag]	Cells rapidly adapt to hyperosmotic stress through coordinated molecular responses. To determine how three-dimensional (3D) chromatin structure contributes to this process, we profiled chromatin interactions, architectural protein occupancy, and transcriptional dynamics in human cells exposed to sorbitol-induced hyperosmotic stress. We performed time-resolved Hi-C to measure stress-induced remodeling of chromatin loops and domains, CUT&Tag to quantify CTCF, RAD21, YAP1, and H3K27ac occupancy at loop anchors,

GSE ID	Title	Summary
		and RNA-seq to capture stress-responsive transcriptional programs. These data reveal global loss of pre-existing chromatin contacts and concurrent formation of de novo, transient loops enriched for retained CTCF and cohesin, as well as transcriptional responses that are temporally layered and largely decoupled from loop remodeling. This dataset provides a resource for investigating how nuclear architecture and transcriptional regulation respond to hyperosmotic stress.
GSE169435	The homeotherm-conserved protein CGGBP1 connects heat stress to chromosomal fusions	We report that heat stress induces chromosomal fusion events in cell culture. We describe the findings first using normal primary fibroblasts [Coriell Cell Repository GM02639] and subsequently apply it to a system of HEK293T stable lines called CT and KD [described in PMID 31547883]. The stable line CT stands for Control shRNA and KD stands for a knockdown of human CGGBP1 using specific shRNA [Origene Lentiviral system TL313955V]. Our results suggest that as measured by the frequency of chromosomal fusions, the CGGBP1 knockdown in KD,

GSE ID	Title	Summary
		when compared to CT, imitates the effects of heat stress. Further, ChIP-sequencing for TP53BP1 show genome-wide misdirected repairs at repeat flanks in KD as compared to CT. We show that CGGBP1 depletion and heat give rise to chromosomal fusions due to overlapping mechanisms that involve misdirected repair in the flanks of repeats genome-wide.
GSE310051	Hyperosmotic stress induces a large-scale rewiring of 3D chromatin interactions [Hi-C]	Cells rapidly adapt to hyperosmotic stress through coordinated molecular responses. To determine how three-dimensional (3D) chromatin structure contributes to this process, we profiled chromatin interactions, architectural protein occupancy, and transcriptional dynamics in human cells exposed to sorbitol-induced hyperosmotic stress. We performed time-resolved Hi-C to measure stress-induced remodeling of chromatin loops and domains, CUT&Tag to quantify CTCF, RAD21, YAP1, and H3K27ac occupancy at loop anchors, and RNA-seq to capture stress-responsive transcriptional programs. These data reveal global loss of pre-existing chromatin contacts and concurrent formation of de novo,

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		transient loops enriched for retained CTCF and cohesin, as well as transcriptional responses that are temporally layered and largely decoupled from loop remodeling. This dataset provides a resource for investigating how nuclear architecture and transcriptional regulation respond to hyperosmotic stress.
GSE124853	ATAC-seq of H1 Human Embryonic Stem Cells (hESC)	ATAC-seq was carried out to identify regions of nucleosome depletion marking sites of enhancers and promoters in the H1 human embryonic stem cell line (H1-hESC)
GSE312288	Hyperosmotic stress induces a large-scale rewiring of 3D chromatin interactions [mAID2 Hi-C]	Cells rapidly adapt to hyperosmotic stress through coordinated molecular responses. To determine how three-dimensional (3D) chromatin structure contributes to this process, we profiled chromatin interactions, architectural protein occupancy, and transcriptional dynamics in human cells exposed to sorbitol-induced hyperosmotic stress. We performed time-resolved Hi-C to measure stress-induced remodeling of chromatin loops and domains, CUT&Tag to quantify CTCF, RAD21, YAP1, and H3K27ac occupancy at loop anchors, and RNA-seq to capture

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		stress-responsive transcriptional programs. These data reveal global loss of pre-existing chromatin contacts and concurrent formation of de novo, transient loops enriched for retained CTCF and cohesin, as well as transcriptional responses that are temporally layered and largely decoupled from loop remodeling. This dataset provides a resource for investigating how nuclear architecture and transcriptional regulation respond to hyperosmotic stress.
GSE277764	Nanopore long-read RNA sequencing in untreated (UN), sodium arsenite (SA) and heat shock (HS) stressed HeLa cells	Recent completion of the telomere-to-telomere (T2T) genome assembly has enabled a comprehensive characterization of pericentromeric Sat, SatII, Sat and centromeric -satellite repeats. Sat DNA constitutes ~1.56% of the genome with a reported localization across 16 chromosomes. The transcription activity of Sat DNA across genome and the sequence of Sat transcripts remained largely unclear. We performed nanopore long-read RNA sequencing (RNA-seq) in untreated (UN), sodium arsenite (SA: 0.1mM, 5 h) and heat shock (HS: 42C, 2 h; 37C, 1 h) stressed HeLa cells to characterize Sat transcripts .

GSE ID	Title	Summary
		Since a portion of Sat transcripts is non-polyadenylated, We performed polyadenylated (poly(A)+) and rRNA-depleted (ribo-) nanopore cDNA long-read RNA-seq.
GSE188797	Chromatin accessibility profiling by ATAC-seq	An updated protocol for ATAC-seq, including wet-lab library preparation and guidelines about sequencing and data analysis.
GSE156339	Stress-induced RNAchromatin interactions promote endothelial dysfunction (iMARGI replicate 2)	Chromatin-associated RNA (caRNA) has been proposed as a type of epigenomic modifier. Here, we test whether environmental stress can induce cellular dysfunction through modulating RNA-chromatin interactions. We induce endothelial cell (EC) dysfunction with high glucose and TNF (H+T), that mimic the common stress in diabetes mellitus. We characterize the H+T-induced changes in gene expression by single cell (sc)RNA-seq, DNA interactions by Hi-C, and RNA-chromatin interactions by iMARGI. H+T induce inter-chromosomal RNA-chromatin interactions, particularly among the super enhancers. To test the causal relationship between H+T-induced RNA-chromatin interactions and the expression of EC dysfunction-related genes, we suppress

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		the LINC00607 RNA. This suppression attenuates the expression of SERPINE1, a critical pro-inflammatory and pro-fibrotic gene. Furthermore, the changes of the co-expression gene network between diabetic and healthy donor-derived ECs corroborate the H+T-induced RNA-chromatin interactions. Taken together, caRNA-mediated dysregulation of gene expression modulates EC dysfunction, a crucial mechanism underlying numerous diseases.
GSE291386	Chromatin accessibility profiling of HSPC-derived erythroblast differentiation [ATAC-seq]	A prime goal of regenerative medicine is to replace dysfunctional cells in the body. To design protocols for producing target cells in the laboratory, one may need to consider exponentially large combinations of culture components. Here, we tested the potential of iteratively approximating the target phenotype by quantifying the distance between chromatin profiles (ATAC-seq) of differentiating cells in vitro and their in-vivo counterparts. We tested this approach on the well-studied generation of erythroblasts from haematopoietic stem cells, evaluating a fixed number of components over two sequential differentiation

GSE ID	Title	Summary
		rounds (8x8 protocols). We found that the most erythroblast-like cells upon the first round yielded the most erythroblast-like cells at the second round, suggesting that greedy selection by chromatin approximation can be a viable optimisation strategy. Furthermore, by targeting transcriptional regulators linked to chromatin regions that were incompletely reprogrammed even after two rounds of differentiation, we could make a data-driven selection of additions to the protocol that further improved erythropoiesis. In future, our methodology can help craft notoriously difficult cells in vitro, such as B cells.
GSE111924	Osmostress induced changes of chromatin architecture and transcription in mammalian cells	This SuperSeries is composed of the SubSeries listed below.
GSE38901	HSF1 drives a transcriptional program distinct from heat shock to support highly malignant human cancers [ChIP-Seq]	Heat-Shock Factor 1 (HSF1), master regulator of the heat-shock response, facilitates malignant transformation, cancer cell survival and proliferation in model systems. The common assumption is that these effects are

GSE ID	Title	Summary
		mediated through regulation of heat-shock protein (HSP) expression. However, the transcriptional network that HSF1 coordinates directly in malignancy and its relationship to the heat-shock response have never been defined. By comparing cells with high and low malignant potential alongside their non-transformed counterparts, we identify an HSF1-regulated transcriptional program specific to highly malignant cells and distinct from heat shock. Cancer-specific genes in this program support oncogenic processes: cell-cycle regulation, signaling, metabolism, adhesion and translation. HSP genes are integral to this program, however, even these genes are uniquely regulated in malignancy. This HSF1 cancer program is active in breast, colon and lung tumors isolated directly from human patients and is strongly associated with metastasis and death. Thus, HSF1 rewires the transcriptome in tumorigenesis, with prognostic and therapeutic implications.

GSE ID	Title	Summary
GSE159190	Occupancy of Integrator subunits on DNA after hyperosmotic stress	ChIP-seq was performed using antibodies against Integrator subunits 11 or 3 to measure the occupancy of these proteins on DNA after hyperosmotic stress.
GSE101803	Oncogenes hijack the stress response machinery in T cell acute lymphoblastic leukemia [GRO-Seq]	Oncogenes NOTCH1 and MYC directly regulate HSF1 and other critical proteins of the stress-response pathway in T-ALL. This GRO-Seq experiment demonstrates that release from NOTCH1 inhibition results in upregulation of HSF1 and other key HSPs.
GSE135355	Stress-induced RNAchromatin interactions promote endothelial dysfunction (iMARGI)	Chromatin-associated RNA (caRNA) has been proposed as a type of epigenomic modifier. Here, we test whether environmental stress can induce cellular dysfunction through modulating RNA-chromatin interactions. We induce endothelial cell (EC) dysfunction with high glucose and TNF (H+T), that mimic the common stress in diabetes mellitus. We characterize the H+T-induced changes in gene expression by single cell (sc)RNA-seq, DNA interactions by Hi-C, and RNA-chromatin interactions by iMARGI. H+T induce inter-chromosomal RNA-chromatin interactions, particularly among the super enhancers. To test the causal relationship between H+T-induced RNA-chromatin

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		interactions and the expression of EC dysfunction-related genes, we suppress the LINC00607 RNA. This suppression attenuates the expression of SERPINE1, a critical pro-inflammatory and pro-fibrotic gene. Furthermore, the changes of the co-expression gene network between diabetic and healthy donor-derived ECs corroborate the H+T-induced RNA-chromatin interactions. Taken together, caRNA-mediated dysregulation of gene expression modulates EC dysfunction, a crucial mechanism underlying numerous diseases.
GSE38912	HSF1 drives a transcriptional program distinct from heat shock to support highly malignant human cancers	This SuperSeries is composed of the SubSeries listed below.
GSE93993	Assay for Transposase-Accessible Chromatin of ATRA treated and wild type HL-60 cell line [ATAC-seq]	These ATAC-seq data were generated to correlate with genomic interaction data in a related Hi-C analysis. Analysis of ATAC-seq data revealed series of overlapping or specific open chromatin peaks in ATRA treated and control cells. Those specific peaks showed fewer overlapping with gene promoters than the overlapping peaks. Key

GSE ID	Title	Summary
		transcription factor for blood cell differentiation GATA2 was found only bind in control specific peaks.
GSE270671	An HSF1-JMJD6-HSP feedback circuit promotes cell adaptation to proteotoxic stress [ChIP-Seq]	Heat Shock Factor 1 (HSF1) is best known as the master transcriptional regulator of the heat-shock response (HSR), a conserved adaptive mechanism critical for protein homeostasis (proteostasis). Combining a genome-wide RNAi library with an HSR reporter, we identified JMJD6 as an essential mediator of HSF1 activity. In follow-up studies, we found that JMJD6 is itself a non-canonical transcriptional target of HSF1 which acts as a critical regulator of proteostasis. In a positive feedback circuit, HSF1 binds and promotes JMJD6 expression, which in turn reduces HSP70 R469 monomethylation to disrupt HSP70-HSF1 repressive complexes resulting in enhanced HSF1 activation. Thus, JMJD6 is intricately wired into the proteostasis network where it plays a critical role for cellular adaptation to proteotoxic stress.
GSE202306	Chromatin accessibility changes after BAF complex disruption in	In order to study the genomic regions undergoing chromatin accessibility changes when BAF chromatin remodelling

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	neuroblastoma cells by ATAC-Seq	complex is inhibited, ATAC-Seq of SK-N-BE(2) neuroblastoma cells was performed after silencing of BAF structural subunits ARID1A and ARID1B.
GSE135354	Stress-induced RNAchromatin interactions promote endothelial dysfunction (Hi-C)	Chromatin-associated RNA (caRNA) has been proposed as a type of epigenomic modifier. Here, we test whether environmental stress can induce cellular dysfunction through modulating RNA-chromatin interactions. We induce endothelial cell (EC) dysfunction with high glucose and TNF (H+T), that mimic the common stress in diabetes mellitus. We characterize the H+T-induced changes in gene expression by single cell (sc)RNA-seq, DNA interactions by Hi-C, and RNA-chromatin interactions by iMARGI. H+T induce inter-chromosomal RNA-chromatin interactions, particularly among the super enhancers. To test the causal relationship between H+T-induced RNA-chromatin interactions and the expression of EC dysfunction-related genes, we suppress the LINC00607 RNA. This suppression attenuates the expression of SERPINE1, a critical pro-inflammatory and pro-fibrotic gene. Furthermore, the changes of the co-expression gene network between diabetic and healthy

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		donor-derived ECs corroborate the H+T-induced RNA-chromatin interactions. Taken together, caRNA-mediated dysregulation of gene expression modulates EC dysfunction, a crucial mechanism underlying numerous diseases.
GSE192370	Genome-wide maps of HSF1 phase separation in heat shock and non-heat shock cells	HSF1 binds DNA via the DBD domain, causing gene upregulation during HS. We assessed the effect of RD-mediated phase separation on chromatin targeting of HSF1 using Cut&Tag followed by high-throughput sequencing to map genome-wide binding of LLPS-competent versus LLPS-incompetent HSF1 mutants under both HS and NHS conditions. Comparing with WT HSF1 under NHS, both WT under HS and M1 under NHS showed increased and broad binding to enhancers and distal intergenic region, with binding most enriched in expected motifs of HSF-related transcription factors. To further assess the role for IDR-induced LLPS in chromatin targeting of HSF1, we used several additional strategies. First, the treatment of 1,6-hexanediol markedly decreased chromatin occupancy both of WT under HS and M1 under NHS conditions. Second, we

GSE ID	Title	Summary
		interrogate chromatin binding of LLPS-deficient mutant M3. Cut&Tag analysis revealed that M3 showed decreased chromatin binding compared to WT under HS and M1 under NHS. The decreased chromatin binding of M3 to chromatin was not due to the loss of its DNA binding ability, as the EMSA assay revealed that M3 was still capable of binding HSE. Instead, the decreased chromatin binding reflects the loss of inter-molecular interaction between HSF1 that holds LLPS. Furthermore, M3 in heat shocked cells shows similar reduced genomic targeting and shallow binding pattern as NHS cells . Third, the enrichment of transcriptional apparatus RNA Pol II, CYCT1, BRD4 to HSF1 target genes also depend on whether HSF1 can phase separate at these sites. Lastly, we conducted live cell single molecule imaging to evaluate chromatin binding kinetics of LLPS-deficient mutant M3 relative to WT HSF1. Measurements of single molecule displacement and diffusion coefficient showed M3 to be significantly more mobile than WT under HS, which suggests M3 was less confined within phase-separated puncta compared

GSE ID	Title	Summary
		with LLPS-competent HSF1. Consistent with this result, super resolution imaging of M3 also showed decreased cluster formation at HSP gene foci but maintained nSBs formation. Altogether, LLPS-forming capability of HSF1 is essential for the efficient recruitment of HSF1 and transcriptional apparatus to HSP gene loci.
GSE112379	Nascent-protein ubiquitination is required for heat shock-induced gene downregulation in human cells.	Proteotoxic stress such as heat shock causes heat-shock factor (HSF)-dependent transcriptional upregulation of chaperones. Heat shock also leads to a rapid and reversible downregulation of many genes, a process we term stress-induced transcriptional attenuation (SITA). The mechanism underlying this conserved phenomenon is unknown. Here we report that enhanced recruitment of negative transcription elongation factors to gene promoters in human cell lines induces SITA. A chemical inhibitor screen showed that active translation and protein ubiquitination are required for the response. We further find that proteins translated during heat shock are subjected to ubiquitination and that p38 kinase signaling connects cytosolic translation with gene downregulation. Notably, brain

GSE ID	Title	Summary
		samples of subjects with Huntington's disease also show transcriptional attenuation, which is recapitulated in cellular models of protein aggregation similar to heat shock. Thus our work identifies an HSF-independent mechanism that links nascent-protein ubiquitination to transcriptional downregulation during heat shock, with potential ramifications in neurodegenerative diseases.
GSE139882	Dynamic regulation of stress-responsive non-genomic CTCF complexes [ChIP-seq]	The nuclear protein CCCTC-binding factor (CTCF) has diverse roles in chromatin architecture and gene regulation. Functionally, CTCF associates with thousands of genomic sites and interacts with proteins, such as cohesin, or non-coding RNAs to facilitate specific transcriptional programming. In this study, we examined CTCF during the cellular stress response in human primary cells using immune-blotting, quantitative real time-PCR, chromatin immunoprecipitation-sequence analysis, mass spectrometry, RNA immunoprecipitation-sequence analysis, and Airyscan confocal microscopy. Unexpectedly, we found that CTCF is exquisitely sensitive to diverse forms of stress in normal patient-derived human

GSE ID	Title	Summary
		mammary epithelial cells (HMECs). In HMECs, the majority of CTCF protein forms non-genomic complexes that localize to Serine/arginine-rich splicing factor (SC-35)-containing nuclear speckles, exclusive of its canonical association with chromatin. Upon stress, non-genomic CTCF protein is rapidly downregulated by changes in protein stability, resulting in loss of CTCF from SC-35 nuclear speckles and changes in CTCF-RNA interactions. CTCF complexes that associate with genomic DNA are resistant to stress-induced degradation and CTCF-DNA binding is largely unchanged. Restoration of cellular CTCF protein abundance and re-localization to nuclear speckles can be achieved by inhibition of proteasome-mediated degradation. Surprisingly, we observed the same characteristics of the stress response during neuronal differentiation of human pluripotent stem cells (hPSC). CTCF forms stress-sensitive complexes that localize to SC-35 nuclear speckles during a specific stage of neuronal commitment/development but not in differentiated neurons. We speculate that these non-canonical CTCF complexes

GSE ID	Title	Summary
		serve a largely non-genomic role in RNA processing, potentially to maintain cells in a particular differentiation state, that is dynamically regulated by environmental signals. The non-canonical, stress-regulated activity of CTCF is uncoupled in persistently stressed, epigenetically re-programmed variant HMECs and certain cancer cell lines. These results reveal new insights into CTCF function in cell differentiation and the stress-response with implications for oxidative damage-induced cancer initiation and neuro-degenerative diseases.
GSE24326	Genome-wide profiling of H3K56 acetylation and transcription factor binding sites in human adipocytes	We use ChIP-seq to discover the genome-wide sites of acetylation of lysine 56 of the histone H3 (H3K56), which is a target of three histone modifying enzymes with known roles in diabetes and insulin resistance, in human adipocytes derived from mesenchymal stem cells. Surprisingly, we find that a very large fraction of genes show some level of acetylation on H3K56, but the highest levels of acetylation are associated with genes previously reported to be involved in type 2 diabetes. Using computational methods, we propose that the

GSE ID	Title	Summary
		transcription factor E2F4 may be involved in recruiting histone modifying enzymes to these sites. We confirm this prediction by measuring the binding of E2F4 using ChIP-seq. We also examine the binding of two other proteins using ChIP-Seq: HSF-1 and C/EBP. HSF-1 is a master regulator of stress responses, and is a target of the same histone modifiers as H3K56. We find a high degree of overlap between HSF-1 binding and H3K56 acetylation even in cells that are not stressed. By contrast, C/EBP, which is not known to be modified by these enzymes, shows much less overlap with the sites of H3K56 acetylation. Our results represent the first mapping of the regulatory code of human adipocytes.
GSE63589	Genome-wide characterization of HSF1 binding in human 451Lu melanoma cancer cells under basal conditions	HSF1 orchestrates a transcriptional program vital for cancer cells. In this study we assayed for genome-wide localization of HSF1 enrichment in the 451Lu melanoma in untreated cells. These results revealed a transcriptional program enriched for metastasis-related genes.
GSE71338	Effects of biobanking on chromatin accessibility	The goal of this study was to test the effects of biobanking on chromatin assessibility, as

GSE ID	Title	Summary
		defined by ATAC-seq. Human nave B cells were FACS sorted from PBMCs that were either frozen or processed fresh. A range of cell numbers were used to identify technical limitations and reproducibility of the ATAC-seq assay on biobanked PBMCs comapred to freshly sorted material. To provide insight into a disease process, naive B cells were isolated from healthy donors or a biorepository of Systemic Lupus Erythematosus undergoing disease flares. These data identify a unique signature of accessible chromatin in SLE B cells.
GSE164200	m6A-RIP-seq of heat shock-exposed HeLa cells upon HSATIII knockdown	To investigate the effect of HSATIII lncRNA on m6A modification, we performed m6A-RIP(RNA immuno precipitation) RNA-seq from heat shock-exposed HeLa cells upon HSATIII knockdown.
GSE72089	Validation of THS-seq method, and comparison of published ENCODE DNase-seq data, self-generated ATAC-seq data and published ATAC-seq data, and THS-seq data for quantitation of chromatin accessibility.	Chromatin accessibility captures the binding status of protein factors to chromosomes in vivo, and has been considered a highly informative proxy for functional protein-DNA interactions. Existing DNase I and Tn5 transposase based assays generally require tens of thousands to millions of fresh cells. Applying Tn5 tagmentation to single cells

GSE ID	Title	Summary
		yields very sparse maps. Here we present a transposome hypersensitive sites sequencing assay (THS-seq) for highly sensitive characterizations of chromatin accessibility.
GSE144303	Dynamic chromatin accessibility landscape changes following interleukin-1 stimulation	Genome-wide methods for examining chromatin modification provide detailed information on regulatory regions of the genome. Dynamic modifications of chromatin allow rapid access of the gene regulatory machinery to condensed genomic regions facilitating subsequent gene expression. Inflammatory cytokine stimulation of cells can cause rapid gene expression changes through direct signalling pathway-mediated transcription factor activation and regulatory element binding.

GSE ID	Title	Summary
GSE222907	ATAC-seq of in vitro differentiated nave, Th1, and Th9 cells	To investigate changes in chromatin accessibility over time, we differentiated Th9 cells in vitro from nave T cells from human and murine sources, and followed them over an extended time course, during which they gained and then lost capacity for IL-9 production
GSE303801	Pre-occupied ATF4 relies on CEBP co-binding for robust transcription activation during integrated stress response [HeLa_CCCP_CUT&Tag]	Upon encountering stress, cells need to rapidly and extensively remodel their transcriptome to restore homeostasis and ensure survival. However, the molecular mechanisms underlying stress-induced transcriptional remodeling are not well understood. In this study, we employed multi-omics approaches to comprehensively profile the changes in transcriptome, epigenetic landscapes, and 3D genome organization during the integrated stress response (ISR), a conserved cellular pathway activated by diverse stress stimuli. Our findings reveal that widespread transcriptional changes occur 4-6 hours after the induction of ISR, coinciding with the genome-wide binding of key transcriptional effector ATF4. Notably, while ATF4 binds to most of its target genes only upon ISR activation, it also

GSE ID	Title	Summary
		occupies the promoters of hundreds of genes under non-stress conditions, priming them for enhanced transcriptional activation in response to stress signals. We further demonstrate that ATF4-mediated gene activation does not rely on traditional transcriptional regulatory mechanisms, such as increased H3K27 acetylation, chromatin accessibility, or remodeling of enhancer-promoter loops. Instead, our data suggest that the nuclear localization of ATF4 induces a pronounced genomic redistribution of transcriptional factor CEBP-, from non-ATF4 sites to a subset of ATF4-bound regions, where it collaborates with ATF4 to elicit transcriptional activation. The combined occupancy of ATF4 and CEBP- ultimately dictates the transcriptional outcomes during ISR. Interestingly, the preferred binding sites of CEBP- in response to stress are not determined by the strength of the CEBP- or ATF4 motifs, but rather by the higher-order chromatin architecture surrounding the binding sites. These findings provide novel insights into the molecular mechanisms that govern the rapid and efficient transcriptional remodeling

GSE ID	Title	Summary
		during ISR, with potential implications for other stress responses.
GSE220026	Single-cell ATAC-seq analysis of hESCs undergoing neuronal differentiation exposed to paracetamol [scATAC-seq]	Genome-wide chromatin opening profiling of hESCs undergoing neuronal differentiation at four distinct time-points. hESCs are differentiated using a neuronal protocol, and day 20 were charachterized. Cells exposed to 100 or 200 M paracetamol from Day 1.
GSE180545	ATAC-seq analysis of HeLa stable cells expressing LAMPS targeting the HS2 enhancer	We describe that light-activated phase separation at the targeted HS2 enhancer increases chromatin accessibility
GSE274408	Heat Shock Transcription Factor 1 Facilitates Liver Cancer Progression by Driving Super-Enhancer-Mediated Transcription of MYCN	HSF1MYCN axis, a transcription-dependent regulatory axis, could be a prognostic indicator and potential therapeutic target in liver cancer.
GSE118189	ATAC-seq data	The immune system depends on a balanced interplay among a variety of specialized cell types transitioning between resting and stimulated states. While epigenome mapping efforts have focused on resting ex vivo immune cells, efforts to characterize stimulation-induced chromatin changes of diverse cell types

GSE ID	Title	Summary
		has been limited. Thus, we collected ATAC-seq and RNA-seq data from immune cell types under resting and stimulated conditions from blood of healthy individuals, and cell types from fetal thymus samples.
GSE167934	Integrating transcription factor abundance with chromatin accessibility in human erythroid lineage commitment	We present InTAC-seq, a method for simultaneous quantification of genome-wide open chromatin and intracellular protein abundance in fixed cells. Using InTAC we directly observe variation in chromatin accessibility and transcription factor motif occupancy driven by differences in transcription factor protein abundance. By purifying bone marrow progenitor cells based on GATA1 protein expression, we establish its role in both functional and epigenetic restriction of erythroid cell identity in human hematopoiesis.
GSE211724	ATAC-seq profiles of glioblastoma stem cells (GSCs), Non-stem tumor cells (NSTCs) and neural progenitor cell (NPC).	Both chromatin accessibility and histone modifications are known to influence transcriptional regulation. We investigated the differences of chromatin accessibility among GSCs, NSTCs and NPC using ATAC-seq (assay for transposase-accessible chromatin with high throughput sequencing).

GSE ID	Title	Summary
GSE274948	Transcriptomic Landscape of Fibroblasts Under IL-1 Stimulation [ATAC-seq]	In this study, we investigated the chromatin accessibility changes in human dermal fibroblasts in response to IL-1 stimulation, a key pro-inflammatory cytokine. Using ATAC-seq, we compared the open chromatin landscapes between control fibroblasts and those treated with IL-1. The data provide insights into the regulatory elements activated during fibroblast differentiation and their potential role in sustaining inflammation within the skin. These findings contribute to our understanding of fibroblast-mediated inflammation in autoimmune diseases like SLE and offer a resource for identifying novel therapeutic targets.
GSE152062	RNA Polymerase II binding profiles of HEK-293T cells exposed to hyperosmotic stress	Anti-Pol II ChIP-seq was performed to assess the redistribution of RNA Polymerase (Pol) II after hyperosmotic stress.
GSE212744	Selective advantage of epigenetically disrupted cancer cells via phenotypic inertia [ATAC-Seq]	Mutations in over one hundred genes, belonging to diverse classes of epigenetic regulators, resulted in significant fitness advantage under nutrient starvation and acidic conditions, suggesting that disruption of epigenetic control, at multiple levels of the regulatory network, promotes broad resistance of

GSE ID	Title	Summary
		cells to stressful environments. To uncover the molecular mechanisms of such phenomenon, and to understand how loss of epigenetic regulators gives advantage during stress, we examined how chromatin accessibility changes in nutrient-deprived conditions using ATAC-seq. ATAC-seq analysis of melanoma cells expressing 4-7sgRNAs targeting members of the PRC2 complex (EED, EZH2, SUZ12), histone H1B (HIST1H1B), the chromatin remodeller SMARCD1 and a non-expressed control gene (TNP2), and grown in either unperturbed conditions (d0) or under glutamine deprivation (d2).
GSE135481	Global landscape mapping of chromatin accessibility in DNA damage-induced senescent human stromal cells	We report the application of an improved assay for transposase-accessible chromatin with high-throughput sequencing in profiling changes of chromatin openning state in senescent human stromal cells. By obtaining over four billion bases of sequence from Tn5 transposase-processed chromosome DNAs, we generated genome-wide chromatin-state maps of human primary stromal cells (with a starting cell number of 50,000). This study provides a

GSE ID	Title	Summary
		framework for the application of ATAC-Seq technique towards spatiotemporal chromatin configuration of human stromal cells upon DNA damage-induced senescence.
GSE274452	Two distinct chromatin modules regulate proinflammatory gene expression [ChIP-seq]	Various mechanisms have been proposed to explain gene activation and co-regulation, including enhancer-promoter interactions via chromatin looping and the enrichment of transcription factors into hubs or condensates. However, these conclusions often stem from analyses of individual loci, and genome-wide studies exploring mechanistic differences in coupled gene expression are lacking. In this study, we dissected the proinflammatory gene expression program induced by TNF in primary human endothelial cells using NGS- and imaging-based techniques. Our findings, enabled by our novel RWireX approach for single-cell ATAC-seq analysis, revealed two distinct regulatory chromatin modules: autonomous links of co-accessibility (ACs) between separated sites, and domains of contiguous co-accessibility (DCs) with increased local transcription factor binding. ACs and DCs associate with different transcriptional bursting kinetics, highlighting

GSE ID	Title	Summary
		the existence of two structurally and functionally distinct regulatory chromatin modules in proinflammatory response. This identification provides a novel mechanistic framework for understanding how cells achieve both rapid and precise control of gene expression.
GSE231486	Genome-wide Transcription Factor binding maps reveal cell-specific changes in the regulatory architecture of human HSPC	This SuperSeries is composed of the SubSeries listed below.
GSE154066	Cell type-specific Chromatin Accessibility Analysis in the Mouse and Human Brain (Human ATAC-seq experiments)	The Assay for Transposase Accessible Chromatin by sequencing (ATAC-seq) is becoming increasing popular in the neuroscience field where chromatin regulation is thought to be involved in neurodevelopment, activity-dependent gene regulation, hormonal and environmental responses, and pathophysiology of neuropsychiatric disorders. The advantages of using this assay include a small amount of material needed, relatively simple and fast protocol, and the ability to capture a range of gene regulatory elements with a single assay. However, with increasing interest in chromatin research, it is an

GSE ID	Title	Summary
		imperative to have feasible, reliable assays that are compatible with a range of neuroscience study designs in both animals and humans. Here we tested three different protocols for neuronal chromatin accessibility analysis, including a varying brain tissue freezing method followed by fluorescent-activated nuclei sorting (FANS) and the ATAC-seq analysis. Our study shows that the cryopreservation method impacts the number of open chromatin regions that can be identified from frozen brain tissue using the cell-type specific ATAC-seq assay. However, we show that all three protocols generate consistent and robust data and enable the identification of functional regulatory elements, promoters and enhancers, in neuronal cells. Our study also implies that the broad biological interpretation of chromatin accessibility data is not significantly affected by the freezing condition. In comparison to the mouse brain analysis, we reveal the additional challenges of doing chromatin analysis on postmortem human brain tissue. However, we also show that these studies are revealing important cell type-

GSE ID	Title	Summary
		specific information about gene regulation in the human brain. Overall, the ATAC-seq coupled with FANS is a powerful method to capture cell-type specific chromatin accessibility information in the mouse and human brain. Our study provides alternative brain preservation methods that generate high quality ATAC-seq data while fitting in different study designs, and further encourages the use of this method to uncover the role of epigenetic (dys)regulation in healthy and malfunctioning brain.
GSE73206	ATAC-Seq on Cohesin mutant HSPCs	Understanding the impact of cohesin mutations on HSPC chromatin structure
GSE174280	Extensive evaluation of ATAC-seq protocols for native or formaldehyde-fixed nuclei	The Assay for Transposase Accessible Chromatin sequencing (ATAC-seq) is an efficient and easy to implement protocol to measure chromatin accessibility that has been widely used in multiple applications studying gene regulation. While several modifications or variants of the protocol have been published since it was first described, there has not yet been an extensive evaluation of the effects of specific protocol choices head-to-head in a consistent experimental

GSE ID	Title	Summary
		setting. In this study, we tested multiple protocol options for major ATAC-seq components (including three reaction buffers, two reaction temperatures, two enzyme sources, and the use of either native or fixed nuclei) in a well-characterized cell line. In addition, the native conditions were tested in a primary sample type (mouse lung tissue) with two different input amounts. In general, native samples yielded more peaks (particularly at loci not overlapping transcription start sites) than fixed samples, and the temperature at which the enzymatic reaction was carried out had a major impact on data quality metrics for both fixed and native nuclei. However, the effect of various conditions tested was not always consistent between the native and fixed samples. For example, the Nextera and Omni buffers were largely interchangeable across all other conditions, while the THS buffer resulted in markedly different profiles in native samples. In-house and commercial enzymes performed similarly. We found that the relationship between commonly used measures of library quality differed across temperature and fixation, and

GSE ID	Title	Summary
		so evaluating multiple metrics in assessing the quality of a sample is recommended. Notably, we also found that these choices can bias the functional class of elements profiled and so we recommend evaluating several formulations in any new experiments. Finally, we hope the ATAC-seq workflow formulated in this study on crosslinked samples will help to profile archival clinical specimens.
GSE105189	ChIP-seq from GM12878 (ENCSR009MBP)	HSF1 ChIP-seq on human GM12878 | For data usage terms and conditions, please refer to [url] and [url]
GSE69304	Genome-wide analysis of ATF4 binding in histidinol-treated cells	Activating Transcription Factor 4 (ATF4) is a transcription factor induced by the integrated stress response (ISR). This experiment is a genome-wide occupancy profiling of ATF4 in human HAP1 cells. HAP1 is a near-haploid human cell line that was derived from KBM-7 cells isolated from a patient with Chronic Myelogenous Leukemia. We induced ATF4 expression by mimicking amino acid starvation with the drug histidinol. We identified peaks of ATF4 binding using three independent antibodies.

GSE ID	Title	Summary
GSE214790	ATAC-seq analysis for GI-MEN DOX-ASCL1 cells and GI-MEN DOX-ASCL1-4TFs cells.	ATAC-seq analysis was performed in an neuroblastoma cell line to analyze chromatin opening of Gi-MEN cells under four different conditions.
GSE139329	Assay for Transposase Accessible Chromatin with high-throughput sequencing (ATAC-seq) revealed genomic chromatin accessibilities change induced by loss of ARID1A in differentiated (day 4) H9 hESCs	We reported loss of ARID1A promoted neurogenesis and inhibited cardiogenesis. Under microscopy, we observed that spontaneously differentiated cells were induced in ARID1A KO H9 hESCs cultured in mTesR medium. We did not know what cells types were. Here ATAC-seq were used to investigate chromatin accessibilities change in differentiated (day 4) WT H9 hESCs and ARID1A KO hESC cells.
GSE134086	Proteome-wide profiling of Transcriptional Machinery on Accessible Chromatin by biotinylated transposon (various cell lines I)	This study introduced a method for directly identifying the transcriptional machinery on accessible chromatin (ATAC-MS) in a proteome scale. The goals of this study are to integrate ATAC-seq with ATAC-MS to investigate the real-time in vivo transcriptional regulatory processes in organisms.
GSE150868	Determine differences in chromatin accessibility in primary human AML samples by ATAC-seq.	ATAC-Seq was performed for 35 AML primary specimens from primary AML cells, followed by a detailed ATAC-Seq pipeline for data processing. We provide both

GSE ID	Title	Summary
		raw files as well as various processed files such as individual and consensus peaks.
GSE47753	Transposition of native chromatin for fast and sensitive epigenomic profiling of open chromatin, DNA-binding proteins and nucleosome position	We describe an assay for transposase-accessible chromatin using sequencing (ATAC-seq), based on direct in vitro transposition of sequencing adaptors into native chromatin, as a rapid and sensitive method for integrative epigenomic analysis. ATAC-seq captures open chromatin sites using a simple two-step protocol with 50050,000 cells and reveals the interplay between genomic locations of open chromatin, DNA-binding proteins, individual nucleosomes and chromatin compaction at nucleotide resolution. We discovered classes of DNA-binding factors that strictly avoided, could tolerate or tended to overlap with the nucleosome. Using ATAC-seq maps of human CD4+ T cells from a proband obtained on consecutive days, we demonstrated the feasibility of analyzing an individuals epigenome on a timescale compatible with clinical decision-making.
GSE209866	ATAC-seq Identification of Chromatin Accessibility in Both	To identify accessible chromatin regions in the human host cells during

GSE ID	Title	Summary
	Human Host Cells and the Parasite Toxoplasma gondii	Toxoplasma parasite infection (uninfected, RH-infected and Pru-infected human foreskin fibroblasts) and in the obligate intracellular parasite Toxoplasma gondii (Type 1 RH strain and Type 2 Pru strain), ATAC-seq was performed.
GSE303800	Pre-occupied ATF4 relies on CEBP co-binding for robust transcription activation during integrated stress response [HeLa_CCCP_ATAC-seq]	Upon encountering stress, cells need to rapidly and extensively remodel their transcriptome to restore homeostasis and ensure survival. However, the molecular mechanisms underlying stress-induced transcriptional remodeling are not well understood. In this study, we employed multi-omics approaches to comprehensively profile the changes in transcriptome, epigenetic landscapes, and 3D genome organization during the integrated stress response (ISR), a conserved cellular pathway activated by diverse stress stimuli. Our findings reveal that widespread transcriptional changes occur 4-6 hours after the induction of ISR, coinciding with the genome-wide binding of key transcriptional effector ATF4. Notably, while ATF4 binds to most of its target genes only upon ISR activation, it also occupies the promoters of hundreds of genes under non-stress conditions, priming

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		them for enhanced transcriptional activation in response to stress signals. We further demonstrate that ATF4-mediated gene activation does not rely on traditional transcriptional regulatory mechanisms, such as increased H3K27 acetylation, chromatin accessibility, or remodeling of enhancer-promoter loops. Instead, our data suggest that the nuclear localization of ATF4 induces a pronounced genomic redistribution of transcriptional factor CEBP-, from non-ATF4 sites to a subset of ATF4-bound regions, where it collaborates with ATF4 to elicit transcriptional activation. The combined occupancy of ATF4 and CEBP- ultimately dictates the transcriptional outcomes during ISR. Interestingly, the preferred binding sites of CEBP- in response to stress are not determined by the strength of the CEBP- or ATF4 motifs, but rather by the higher-order chromatin architecture surrounding the binding sites. These findings provide novel insights into the molecular mechanisms that govern the rapid and efficient transcriptional remodeling during ISR, with potential implications for other stress responses.

GSE ID	Title	Summary
GSE208030	ATAC-seq(Assay of Transposase Accessible Chromatin with High-throughput sequencing) after ER stress	Endoplasmic reticulum stress was induced by tunicamycin in hepatocellular carcinoma cells, and chromatin opening detection was performed to search for different chromatin opening sites and different target genes, and then to find the different motif.
GSE159724	Heat Shock Factor 1 (HSF1) supports the ESR1 action in breast cancer (ChIP-seq)	Heat shock factor 1 (HSF1) is a key regulator of transcriptional responses to proteotoxic stress. It has been recently linked to signaling of estrogen via ESR1. To study the cooperation of HSF1 and ESR1 in the transcriptional response to estrogen, we established estrogen receptor (ER)-positive breast cancer cell lines with reduced HSF1 levels using specific shRNA or CRISPR/Cas9 approach. HSF1 deficiency led to the inhibition of the mitogenic effect of estrogen in MCF7 and T47D cells. RNA-seq analyses revealed that the stimulatory effect of E2 on the transcriptome was smaller in HSF1-deficient MCF7 cells. This could partially result from the higher basal expression of E2-dependent genes in these cells as a consequence of the enhanced binding of unliganded ESR1 to chromatin, which was revealed by ChIP-seq analyses. Thus, we

GSE ID	Title	Summary
		postulate that some fraction of ESR1 could be released from the inhibitory complex with HSP90 and gain transcriptional competence without E2-stimulation.
GSE281808	Enhancing single-cell ATAC sequencing with formaldehyde fixation, cryopreservation, and multiplexing for flexible analysis [ATAC-seq]	The assay for transposase-accessible chromatin using sequencing (ATAC-seq) revolutionized the field of epigenetics since its emergence by providing a means to uncover chromatin dynamics and other factors affecting gene expression. The development of single-cell (sc) applications in recent years led to an even deeper understanding of cell type specific gene regulatory mechanisms. One of the major challenges while running ATAC-seq experiments, bulk or sc, is the need for freshly collected cells for successful experiments. While various freezing methods have already been tested and established for bulk and sc ATAC-seq, quality metrics for preserved cells are rather poor or dependent on sampling time when compared to fresh samples. This makes it difficult to conduct all sorts of complex experiments i.e. with multiple conditions, patients, or time course studies. Especially, accounting for batch effects can be difficult if

GSE ID	Title	Summary
		samples need to be processed at different time points of collection. We tackled this issue by adding a fixation step prior to the freezing method. The additional fixation step improved library quality and yield data comparable to fresh samples. The workflow was also tested on multiplexed sc ATAC experiments, set-up for cost-efficient low input sample handling. Sample cross-in, typically encountered in Tn5-based multiplex approaches, were tackled with a computational procedure specifically developed for this approach.
GSE93995	Genome-wide Studies of Chromatin Structure during Blood Cell Differentiation [HiC-seq]	Chromatin structure plays key role in regulating whole genome gene expression.The goal of this study are to compare chromatin structure of ATRA treated and control cells, finding the mechanism that how chromatin structure regulates gene transcription.
GSE130450	Quantification, dynamic visualization, and validation of bias in ATAC-seq data with ataqv	The assay for transposase-accessible chromatin using sequencing (ATAC-seq) has become the preferred method for mapping chromatin accessibility, due to its simplicity, speed, and low input material requirements. However, it can be difficult to evaluate data quality, particularly when processing

GSE ID	Title	Summary
		many samples. Here, we present ataqv, a computational toolkit for efficiently measuring, visualizing, and comparing ATAC-seq quality control results. We perform controlled ATAC-seq experiments and utilize ataqv to show that the ratio of Tn5 transposase to nuclei and sequencing flowcell cluster density induce reproducible technical bias in ATAC-seq results, significantly changing the fragment length distribution and enrichment of reads in promoter and enhancer chromatin states.
GSE193644	ATAC-seq time course analysis of in vitro differentiated antibody-secreting cells	We examined the chromatin accessibility changes that occur as antibody-secreting cells (ASC) differentiate and mature using an in vitro culture system. ASC were profiled directly post isolation and at time points weekly out to 14 days in culture. These data reveal how the long-lived ASC accessible chromatin matures over time.
GSE230586	ATAC-seq analysis for T-ALL cell lines	ATAC-seq analysis was performed in multiple T-ALL cell lines to analyze chromatin accessibility.
GSE281809	Enhancing single-cell ATAC sequencing with formaldehyde fixation,	The assay for transposase-accessible chromatin using sequencing (ATAC-seq)

GSE ID	Title	Summary
	cryopreservation, and multiplexing for flexible analysis [scATAC-seq]	revolutionized the field of epigenetics since its emergence by providing a means to uncover chromatin dynamics and other factors affecting gene expression. The development of single-cell (sc) applications in recent years led to an even deeper understanding of cell type specific gene regulatory mechanisms. One of the major challenges while running ATAC-seq experiments, bulk or sc, is the need for freshly collected cells for successful experiments. While various freezing methods have already been tested and established for bulk and sc ATAC-seq, quality metrics for preserved cells are rather poor or dependent on sampling time when compared to fresh samples. This makes it difficult to conduct all sorts of complex experiments i.e. with multiple conditions, patients, or time course studies. Especially, accounting for batch effects can be difficult if samples need to be processed at different time points of collection. We tackled this issue by adding a fixation step prior to the freezing method. The additional fixation step improved library quality and yield data comparable to fresh samples. The workflow was

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		also tested on multiplexed sc ATAC experiments, set-up for cost-efficient low input sample handling. Sample cross-in, typically encountered in Tn5-based multiplex approaches, were tackled with a computational procedure specifically developed for this approach.
GSE128390	Histone Modifications and Chromatin Accessibility in Endothelial Cells under Shear Stress (ATAC-seq)	This data set reveals the changes of histone modifications and chromatin accessibility in human umbilical vein endothelial cells (HUVECs) under atheroprotective pulsatile shear (PS), atheroprone oscillatory shear (OS), or with KLF4 overexpression. Using ChIP-Seq, we defined the H3K27ac and H3K4me1 enrichment under PS and OS conditions. Using ATAC-seq, we identified the chromatin accessibility under KLF4 overexpression.
GSE231857	Dynamic genome Chromatin accessibility of T lymphocyte under microgravity condition	In space flight,microgravity may be sensed and transmitted through cellular and nuclear structures, thereby altering nuclear plasticity, chromatin organization and openness, which in turn leads to differential gene expression. Studies on genome-wide chromatin openness under microgravity environments will help to clarify whether the

GSE ID	Title	Summary
		Earth's gravity is stable and maintains the structure and openness of chromatin, as well as the steady-state conditions that constitute genome modification. At the same time, it will provide a strong scientific basis for astronaut's protection under extreme conditions.
GSE140934	Chromatin Remodeling and Immediate Early Gene Activation by SLFN11 in Response to Replication Stress [ATAC-seq_K562]	Schlafen 11 (SLFN11) has recently arisen as a novel cellular restriction factor against replication stress. Here we show that SLFN11 increases chromatin accessibility genome-wide, dominantly at promoters in response to replication stress induced by the CHK1 inhibitor prexasertib and the topoisomerase I inhibitor, camptothecin. Concomitantly SLFN11 selectively activates cellular stress response pathways by inducing the transcription of the Immediate Early Genes (IEGs) including JUN, FOS, EGR1, NFKB2 and ATF3. Both chromatin opening and IEG activation require the putative helicase activity of SLFN11 whereas extrinsic IEG activation by serum induction is SLFN11-independent. SLFN11-dependent IEG activation is also observed across 55 non-isogenic NCI-60 cell lines by comparing transcriptome data before and

GSE ID	Title	Summary
		after camptothecin treatment. We conclude that SLFN11 acts as a global regulator of chromatin structure and an intrinsic IEG activator with the potential to engage the native immune activation in response to replicative stress.
GSE167298	ATAC-seq in HUDEP-2 cells	ATAC-seq was performed in HUDEP-2 cells grown in expansion phase medium following the OMNI-ATAC protocol (Corces MR, et al. Nat. Methods, 2017).
GSE200269	PRO-IP-seq: Pol II CTD code at nt-resolution in human K562 cells	Active sites of transcription were labelled with biotinylated nucleotide during a run-on reacation, where after an immunoprecipitation with antibodies that target Pol II CTD was conducted. This PRO-IP-seq protocol was conducted in human K562 erythroleukemia cells. The K562 cells were either untreated (NHS) or treated with 30 minutes of heat shock at 42C (HS).
GSE52924	ChIP-seq analysis of chromatin state plasticity in primary human adult erythroid progenitor cells	We report a computational approach for investigation of chromatin state plasticity. We applied this approach to investigate an ENCODE ChIP-seq dataset profiling the genome-wide distribution of H3K27me3 in 19 human cell lines. We found that high plasticity regions (HPRs) can

GSE ID	Title	Summary
		be divided into two functionally and mechanistically distinct groups, consisting of CpG island proximal and distal regions. We identified cell-type specific regulators correlating with H3K27me3 patterns at distal HPRs in ENCODE cell lines. Furthermore, we applied this approach to investigate mechanisms for poised enhancer establishment in primary human erythroid precursors. We predicted and validated a previously unrecognized role of TAL1 in modulating H3K27me3 patterns through interaction with additional cofactors, such as GFI1B. Our integrative approach provides mechanistic insights into chromatin state plasticity and is broadly applicable to other epigenetic marks.
GSE66386	High-resolution nucleosome positioning from ATAC-seq chromatin accessibility data	We describe the NucleoATAC algorithm for high-resolution nucleosome positioning and occupancy determination using ATAC-seq.
GSE273428	Two distinct chromatin modules regulate proinflammatory gene expression [snATAC]	Various mechanisms have been proposed to explain gene activation and co-regulation, including enhancer-promoter interactions via chromatin looping and the enrichment of transcription factors into hubs or condensates. However, these conclusions often stem

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		from analyses of individual loci, and genome-wide studies exploring mechanistic differences in coupled gene expression are lacking. In this study, we dissected the proinflammatory gene expression program induced by TNF in primary human endothelial cells using NGS- and imaging-based techniques. Our findings, enabled by our novel RWireX approach for single-cell ATAC-seq analysis, revealed two distinct regulatory chromatin modules: autonomous links of co-accessibility (ACs) between separated sites, and domains of contiguous co-accessibility (DCs) with increased local transcription factor binding. ACs and DCs associate with different transcriptional bursting kinetics, highlighting the existence of two structurally and functionally distinct regulatory chromatin modules in proinflammatory response. This identification provides a novel mechanistic framework for understanding how cells achieve both rapid and precise control of gene expression.

GSE ID	Title	Summary
GSE128389	Histone Modifications and Chromatin Accessibility in Endothelial Cells under Shear Stress (ChIP-seq)	This data set reveals the changes of histone modifications and chromatin accessibility in human umbilical vein endothelial cells (HUVECs) under atheroprotective pulsatile shear (PS), atheroprone oscillatory shear (OS), or with KLF4 overexpression. Using ChIP-Seq, we defined the H3K27ac and H3K4me1 enrichment under PS and OS conditions. Using ATAC-seq, we identified the chromatin accessibility under KLF4 overexpression.
GSE172005	ATAC-seq of uncultured CD112high and CD112low long-term(LT) human hematopoietic stem cells (HSC) from umbilical cord blood	We performed ATAC-sequencing to elucidate the chromatin accessibility that underlie the differential in vitro priming and regeneration phenotypes of CD112high and CD112low LT-HSC. Therefore, we prospectively isolated these subpopulations from 3 independent human umbilical cord blood pools and subjected to ATAC-sequencing directly. We show that distinct subsets of human LT-HSC respond differently to regeneration-mediated stress and that INKA1 governs these distinct stemness states where CD112low LT-HSC are marked by an alternative state of quiescence with high INKA1 and low H4K16ac preserving self-renewal and regenerative

GSE ID	Title	Summary
		capacity upon successive rounds of regeneration-mediated stress.
GSE273429	Two distinct chromatin modules regulate proinflammatory gene expression [snMultiome]	Various mechanisms have been proposed to explain gene activation and co-regulation, including enhancer-promoter interactions via chromatin looping and the enrichment of transcription factors into hubs or condensates. However, these conclusions often stem from analyses of individual loci, and genome-wide studies exploring mechanistic differences in coupled gene expression are lacking. In this study, we dissected the proinflammatory gene expression program induced by TNF in primary human endothelial cells using NGS- and imaging-based techniques. Our findings, enabled by our novel RWireX approach for single-cell ATAC-seq analysis, revealed two distinct regulatory chromatin modules: autonomous links of co-accessibility (ACs) between separated sites, and domains of contiguous co-accessibility (DCs) with increased local transcription factor binding. ACs and DCs associate with different transcriptional bursting kinetics, highlighting the existence of two structurally and functionally distinct regulatory chromatin

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		modules in proinflammatory response. This identification provides a novel mechanistic framework for understanding how cells achieve both rapid and precise control of gene expression.
GSE100750	Repression of stress-induced LINE-1 expression protects cancer cell populations from lethal drug-exposures [ATAC-Seq]	DTP heterochromatin in genomic repeat regions protects the population from drug-induced death.
GSE33213	Open Chromatin TFBS by ChIP-seq from ENCODE/Open Chrom(UT Austin)	This data was generated by ENCODE. If you have questions about the data, contact the submitting laboratory directly (Terry Furey mailto:tsfurey@duke.edu). If you have questions about the Genome Browser track associated with this data, contact ENCODE (mailto:genome@soe.ucsc.edu). | These tracks display a synthesis of evidence from different assays as part of the four Open Chromatin track sets. This track displays open chromatin regions and/or transcription factor binding sites identified in multiple cell types by one or more complementary methodologies, DNaseI hypersensitivity (HS) (Duke DNaseI HS), Formaldehyde-Assisted Isolation of Regulatory Elements (FAIRE)

GSE ID	Title	Summary
		(UNC FAIRE), and chromatin immunoprecipitation (ChIP) for select regulatory factors (UTA TFBS). Each methodology was performed on the same cell type using identical growth conditions. (Note: Data for some or all ChIP experiments may not be available for all cell types). Regions that overlap between methodologies identify regulatory elements that are cross-validated indicating high confidence regions. In addition, multiple lines of evidence suggest that regions detected by a single assay (e.g., DNase-only or FAIRE-only) are also biologically relevant (Song et al., submitted). | DNaseI HS data: DNaseI is an enzyme that has long been used to map general chromatin accessibility, and DNaseI hypersensitivity is a feature of active cis-regulatory sequences.The use of this method has led to the discovery of functional regulatory elements that include promoters, enhancers, silencers, insulators, locus control regions, and novel elements. DNaseI hypersensitivity signifies chromatin accessibility following binding of trans-acting factors in place of a canonical nucleosome. | FAIRE data: FAIRE (Formaldehyde

GSE ID	Title	Summary
		Assisted Isolation of Regulatory Elements) is a method to isolate and identify nucleosome-depleted regions of the genome. FAIRE was initially discovered in yeast and subsequently shown to identify active regulatory elements in human cells (Giresi et al., 2007). Similar to DNaseI HS, FAIRE appears to identify functional regulatory elements that include promoters, enhancers, silencers, insulators, locus control regions and novel elements. | ChIP data: ChIP (Chromatin Immunoprecipitation) is a method to identify the specific location of proteins that are directly or indirectly bound to genomic DNA. By identifying the binding location of sequence-specific transcription factors, general transcription machinery components, and chromatin factors, ChIP can help in the functional annotation of the open chromatin regions identified by DNaseI HS mapping and FAIRE. | Input data: As a background control experiment, we sequenced the input genomic DNA sample that was used for ChIP. Crosslinked chromatin is sheared and the crosslinks are reversed without carrying out the immunoprecipitation step.

GSE ID	Title	Summary
		This sample is otherwise processed in a manner identical to the ChIP sample as described below. The input track is useful in revealing potential artifacts arising from the sequence alignment process such as copy number differences between the reference genome and the sequenced samples, as well as regions of poor sequence alignability. | For data usage terms and conditions, please refer to [url] and [url]
GSE125212	ATAC-seq of immunophenotypic LT-HSC and ST-HSC from umblical cord blood in cytokine-rich ex vivo culture conditions following sphingolipid modulation	To investigate how ex vivo culture affects chromatin accessibility in cultured HSC, we performed the Assay for Transposase Accessible Chromatin with high-throughput sequencing (ATAC-Seq) on cLT (CD34+CD90+CD45RA-) and cST populations purified from 8 day cultured lineage depleted cord blood (lin- CB) cells treated with 3-Factor (4HPR+UM171+SR1), U+S or 4HPR as well as untreated and vehicle-treated (DMSO) control populations. The subsequent ATAC-seq data was compared to chromatin accessibility signatures generated from uncultured hematopoietic stem and progenitor populations (Takayama, et al.). We found that ex vivo culture shifted cLT and cST cells

GSE ID	Title	Summary
		isolated from control or untreated samples to a chromatin accessibility profiles not found in LT-HSC, suggesting some loss of a stem-cell associated chromatin state. By contrast, 4HPR-treated, to some extent, and 3-Factor-treated HSC maintained chromatin accessibility features of uncultured LT-HSC.